

ZHUMADIAN, ChinaWMO: **572900**Lat: **32.9403N**Lon: **113.9144E**Elev: **107**StdP: **100.04**Time Zone: **8.00 (E08)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.0	-3.6	-17.0	0.9	0.9	-14.0	1.1	1.5	7.4	5.8	6.4	5.4	1.3	270	0.370

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.2	35.5	25.6	34.2	25.2	33.0	24.9	28.2	33.0	27.5	32.0	26.8	31.0	3.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
27.1	23.1	31.1	26.3	22.0	30.4	25.6	21.2	29.7	92.2	33.0	88.6	32.1	85.4	31.0	30.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.3	5.3	4.6	-8.3	37.6	2.3	1.3	-10.0	38.6	-11.3	39.3	-12.6	40.1	-14.3	41.0		
(5)			-9.4	29.3	2.1	0.8	-10.9	29.9	-12.2	30.3	-13.3	30.8	-14.8	31.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.0	2.1	5.0	10.6	16.5	21.8	26.1	27.8	26.6	22.3	17.4	10.5	4.4
	DBStd	9.53	3.29	3.95	4.39	4.19	3.44	2.93	2.71	2.93	3.09	3.58	4.15	3.33
	HDD10.0	658	247	146	45	4	0	0	0	0	0	1	42	174
	HDD18.3	1944	504	374	242	82	9	1	0	0	5	61	237	431
	CDD10.0	2846	1	5	64	199	367	484	551	516	369	230	56	2
	CDD18.3	1088	0	0	2	28	117	234	293	258	124	32	1	0
	CDH23.3	10858	0	0	19	244	1041	2508	3337	2614	898	188	7	0
CDH26.7	4494	0	0	3	61	380	1129	1516	1071	295	38	1	0	
Wind	WSAvg	2.1	2.0	2.3	2.5	2.3	2.2	2.2	2.0	1.7	1.7	1.8	2.0	2.0
Precipitation	PrecAvg	831	18	25	40	55	83	107	184	133	83	51	37	16
	PrecMax	1140	59	67	81	105	213	218	341	257	189	133	96	53
	PrecMin	576	3	3	6	12	16	46	66	37	11	5	1	1
	PrecStd	131	14	14	22	24	40	39	69	51	47	39	27	14
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.7	19.1	25.5	31.2	34.7	36.9	36.7	36.8	34.4	30.1	24.4	16.7
		MCWB	7.5	11.9	16.7	20.6	20.4	22.8	27.4	27.7	24.4	18.8	14.9	8.9
	2%	DB	11.7	15.9	22.8	28.2	32.4	35.1	35.4	34.4	31.8	27.4	21.6	13.8
		MCWB	5.8	9.2	14.8	19.0	20.1	23.0	27.2	27.1	23.1	17.4	13.4	7.3
	5%	DB	9.6	13.6	20.3	26.2	30.6	33.7	34.2	33.0	29.8	25.4	19.1	11.6
		MCWB	4.5	7.8	13.4	18.0	20.1	22.9	27.0	26.5	21.9	16.7	12.4	6.1
	10%	DB	7.6	11.6	17.9	23.9	28.8	32.2	32.9	31.6	28.1	23.3	16.9	9.6
		MCWB	3.4	6.8	11.8	16.6	19.6	22.5	26.4	25.7	21.0	16.1	11.3	5.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.5	12.8	18.3	22.4	24.0	27.2	29.2	29.2	26.9	21.2	16.5	9.9
		MCDB	13.0	17.4	23.6	27.9	29.4	32.7	34.0	33.6	32.6	24.7	21.9	15.1
	2%	WB	6.4	10.3	15.9	20.4	22.8	26.1	28.5	28.2	25.2	19.5	14.7	8.4
		MCDB	10.8	14.5	21.4	26.2	28.2	31.4	33.3	32.7	29.0	24.1	19.7	12.1
	5%	WB	5.1	8.7	14.0	18.9	21.8	25.2	27.8	27.5	23.7	18.4	13.3	7.1
		MCDB	8.9	12.2	19.1	24.2	27.5	30.3	32.5	31.5	27.2	22.9	17.6	10.4
	10%	WB	3.8	7.5	12.4	17.6	20.8	24.3	27.1	26.7	22.5	17.4	11.9	5.9
		MCDB	6.9	11.0	17.1	22.8	26.5	29.3	31.6	30.3	25.9	21.6	15.8	8.9
Mean Daily Temperature Range	5% DB	MDBR	7.3	8.0	8.7	9.6	9.7	9.0	7.2	7.1	8.3	8.3	7.9	7.2
		MCDBR	11.3	11.7	12.3	13.0	12.8	11.5	9.1	8.9	11.2	11.8	12.0	11.4
		MCWBR	6.9	6.8	6.8	6.7	4.8	3.6	3.2	3.3	3.9	4.7	5.8	6.6
	5% WB	MCDBR	10.4	10.1	11.5	11.5	10.3	8.7	8.0	7.8	8.7	9.4	10.2	9.7
		MCWBR	6.7	6.4	7.0	6.8	5.0	3.5	3.5	3.4	4.0	4.4	5.6	6.0
Clear-Sky Solar Irradiance	taub	0.623	0.681	0.685	0.678	0.685	0.760	0.724	0.750	0.717	0.711	0.667	0.612	
	taud	1.566	1.498	1.512	1.559	1.593	1.494	1.596	1.534	1.560	1.514	1.548	1.603	
	Ebn at Noon	553	576	626	661	664	614	634	606	597	546	514	530	
	Edh at Noon	223	262	277	275	268	295	265	278	261	254	225	204	
All-Sky Solar Radiation	RadAvg	2.17	2.56	3.76	4.48	4.77	4.78	4.53	4.24	3.55	2.94	2.38	2.08	
	RadStd	0.30	0.53	0.37	0.42	0.39	0.43	0.58	0.61	0.53	0.40	0.36	0.35	